

Linear Algebra and its Applications (David C Lay)

1.1 Systems of Linear Equations

A linear equation in the variables $x_1, \dots, x_n$ is an equation that can be written in the form $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$ (1) where b and the coefficients $a_1, \dots, a_n$ are real or complex numbers, usually known in advance. The subscript n may be any positive integer.

The equations $4x_1 - 5x_2 + 2 = x_1$ and $x_2 = 2(\sqrt{6} - x_1) + x_3$ are both linear because they can be rearranged algebraically as in equation (1): $3x_1 - 5x_2 = -2$ and $2x_1 + x_2 - x_3 = 2\sqrt{6}$

The equations $4x_1 - 5x_2 = x_1x_2$ and $x_2 = 2\sqrt{x_1} - 6$ are not linear because of the presence of x_1x_2 in the first equation and $\sqrt{x_1}$ in the second.

A system of linear equations (or a linear system) is a collection of one or more linear equations involving the same variables — say, $x_1, \dots, x_n$.

An example is
$$\begin{cases} 2x_1 - x_2 + 1.5x_3 = 8 \\ x_1 - 4x_3 = -7 \end{cases} \quad (2)$$